

Performance Constrained Greedy Selection and Supermodular Approximation under Quadratic Observations

Wooyeong Cho, Tengyou Xu, Kaden Bautista and Ankur Mehta

Abstract—We address the problem of selecting the smallest set of measurements for nonlinear quadratic observation models under a prescribed estimation-accuracy constraint. This problem is NP-hard and challenging to solve because exact optimization requires combinatorial search. Moreover, the posterior covariance under quadratic observations is generally intractable, and the resulting objective lacks exact supermodularity, limiting the direct use of existing greedy guarantees. To address this challenge, we optimize a closed-form Van Trees surrogate that remains explicitly connected to estimation performance and analyze greedy measurement selection on the resulting A-optimality objective through the supermodularity ratio. Our main theoretical result establishes a measurement-utilization guarantee relative to the optimal subset size and derives an explicit lower bound on the supermodularity ratio for a substantially broader class of quadratic observation models than prior work, including those with nonzero linear terms and full-rank quadratic sensing matrices.

I. INTRODUCTION

Optimal sensor selection is the problem of choosing a subset of available sensors or measurements to achieve a desired estimation or monitoring performance while respecting practical resource constraints such as limited energy, communication bandwidth, and computation. This problem arises in many real-world systems, including autonomous robots [1], [2], distributed sensor networks [3], [4], intelligent transportation systems [5], and surveillance or target-tracking applications [6], [7], where large numbers of sensors are available but only a fraction can be activated efficiently at any given time. In such settings, continuously using every sensor is often costly or even infeasible. Consequently, the goal of optimal sensor selection is to identify the smallest or most informative subset of measurements that preserves the required performance while reducing overall resource usage.

A fundamental challenge in optimal sensor selection is its combinatorial nature. As the number of available sensors increases, the number of possible measurement subsets grows exponentially, making exact optimization computationally intractable in practice. Existing work [8], [9], [10] has identified this combinatorial complexity as a central obstacle and has established theoretical limits as well as exact formulations for optimal selection.

Because the problem is NP-hard [10], [11], a major line of prior research has focused on identifying structural properties of the objective that enable efficient approximation

algorithms. In particular, submodularity and supermodularity have played an important role in justifying greedy selection methods with provable performance guarantees [12], [13], [14]. Even when the objective does not satisfy these properties exactly, approximate or weak submodularity/supermodularity has been used to extend greedy guarantees to more general settings. For example, [15], [16] leverage weak supermodularity to analyze minimization of non-submodular estimation metrics such as mean-square error. These works show that even when the objective does not exhibit exact diminishing returns, suitable constant-factor characterizations can still quantify its deviation from ideal structure and yield meaningful greedy performance guarantees.

Beyond the combinatorial challenge, many practical sensing problems introduce an additional difficulty: nonlinear observation models [17], [18]. In these settings, the estimation objective is often no longer available in closed form, which complicates both analysis and optimization. To address this issue, [19] used the Van Trees bound as a principled surrogate that remains explicitly connected to estimation performance while preserving analytical tractability under quadratic observation models.

However, the existing analysis in [19] applies to a restrictive special case of quadratic observations. In particular, it assumes a pure quadratic model without linear term and rank-1 quadratic observation factors. As a result, the corresponding guarantees do not directly extend to more general sensing models with mixed linear-quadratic structure or full-rank quadratic terms. This leaves an unresolved gap between the available theory and the broader class of nonlinear observation models encountered in practice.

In this paper, we address this gap by formulating a performance-constrained sensor selection problem for a broader class of nonlinear quadratic observation models. Specifically, we study the resource-aware problem of achieving a prescribed estimation-accuracy level with the smallest possible number of selected measurements. To solve this problem, we optimize a Van Trees-based A-optimality surrogate and develop a greedy approximation framework for the resulting non-supermodular objective. Our analysis establishes a corresponding measurement-utilization guarantee and, unlike prior work, derives a lower bound on the supermodularity ratio that remains valid for models with nonzero linear terms and full-rank quadratic sensing matrices. In this way, our work not only extends the structural analysis of quadratic observation models, but also connects that analysis to an explicit bound on sensing resource usage.

Contributions. The main contributions of this paper are summarized as follows:

- 1) Formulate a performance-constrained minimum-cardinality sensor selection problem for nonlinear quadratic observation models.
- 2) Develop a greedy approximation framework for this non-supermodular objective and derive a corresponding measurement-utilization guarantee.
- 3) Establish explicit lower bound on the supermodularity ratio of the proposed set function.
- 4) Demonstrate that the bound remains valid by simulation results, especially for observation models with nonzero linear terms and full-rank quadratic factors.

II. PRELIMINARIES

Notations: $\mathbb{R}$ denotes the set of real numbers, and $\mathbb{Z}_+$ represents the set of non-negative integers. $\text{Tr}(\cdot)$ is the trace operator, which returns the sum of the diagonal elements of a square matrix. $X^\top$ is the transpose of X and X^{-1} represents the inverse of X .

Lemma 1. For multiplicatively compatible matrices $A_1, A_2, \dots, A_m$, $\text{Tr}(A_1 A_2 \cdots A_m) = \text{Tr}(A_m A_1 \cdots A_{m-1})$.

Lemma 2. If $A, B \in \mathbb{R}^{n \times n}$ are symmetric positive definite and $A \preceq B$, then $B^{-1} \preceq A^{-1}$.

Lemma 3. If $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite, then

$$\lambda_{\min}(A^{-1}) = \lambda_{\max}(A)^{-1}, \quad \lambda_{\max}(A^{-1}) = \lambda_{\min}(A)^{-1}.$$

Lemma 4. If $a, b \geq 0$ and $c, d > 0$, then

$$\frac{a+b}{c+d} \geq \min \left\{ \frac{a}{c}, \frac{b}{d} \right\}.$$

Lemma 5. If $A \in \mathbb{R}^{n \times n}$ is symmetric positive semidefinite, then for all $x \in \mathbb{R}^n$, $\lambda_{\min}(A) \|x\|^2 \leq x^\top A x \leq \lambda_{\max}(A) \|x\|^2$.

Lemma 6. If $A, B \in \mathbb{R}^{n \times n}$ are symmetric positive semidefinite, then $\lambda_{\min}(A) \text{Tr}(B) \leq \text{Tr}(AB) \leq \lambda_{\max}(A) \text{Tr}(B)$.

Lemma 7. Let $A \in \mathbb{R}^{n \times n}$ and $C \in \mathbb{R}^{k \times k}$ be invertible, and let $U \in \mathbb{R}^{n \times k}$, $V \in \mathbb{R}^{k \times n}$. Then

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}.$$

A. Set functions

Let G be a ground set. A set function f over G assigns a real number to each subset of G , i.e., $f: 2^G \rightarrow \mathbb{R}$.

Definition 1. A set function $f: 2^G \rightarrow \mathbb{R}$ is *monotone decreasing* if and only if for any sets $S_1, S_2 \subseteq G$ satisfying $S_1 \subseteq S_2$,

$$f(S_1) \geq f(S_2).$$

Definition 2. Let $f: 2^G \rightarrow \mathbb{R}$ be a set function defined on a finite ground set G . We say that f is *supermodular* if for any sets $S_1 \subseteq S_2 \subseteq G$ and any element $j \in G \setminus S_2$,

$$f(S_1) - f(S_1 \cup \{j\}) \geq f(S_2) - f(S_2 \cup \{j\}).$$

In other words, as the selected set grows from S_1 to S_2 , the decrease caused by adding the same element j becomes smaller.

Definition 3. Let $f: 2^G \rightarrow \mathbb{R}$ be a nonincreasing set function on a finite ground set G . The *supermodularity ratio* of f is defined as

$$\gamma_f := \min_{S_1 \subseteq S_2 \subseteq G, j \in G \setminus S_2} \frac{f(S_1) - f(S_1 \cup \{j\})}{f(S_2) - f(S_2 \cup \{j\})}.$$

The quantity γ_f measures how closely f follows the supermodularity property. In particular, $\gamma_f \in [0, 1]$. When $\gamma_f = 1$, the function is supermodular. When $0 < \gamma_f < 1$, the function may not be exactly supermodular, but it still satisfies the relaxed inequality

$$f(S_1) - f(S_1 \cup \{j\}) \geq \gamma_f (f(S_2) - f(S_2 \cup \{j\})),$$

for all $S_1 \subseteq S_2 \subseteq G$ and $v \in G \setminus S_2$. Hence, γ_f quantifies the degree to which f is approximately supermodular.

III. PROBLEM FORMULATION

A. System description

$$y_t = \frac{1}{2} \sum_{i=1}^m e^{(i)} (x_t^\top M^{(i)} x_t) + Cx_t + v_t \quad (1)$$

The system measurement at time t is denoted by $y_t \in \mathbb{R}^m$. The above equation defines a nonlinear quadratic measurement model composed of a linear term and a quadratic term in the state. The linear term is given by the measurement matrix $C \in \mathbb{R}^{m \times n}$, which maps the state directly into the measurement space. The quadratic term captures pairwise interactions among the state components. Specifically, the i -th quadratic measurement channel is written as $x_t^\top M^{(i)} x_t$, where $M^{(i)} \in \mathbb{R}^{n \times n}$ is a known matrix. The standard basis vector $e^{(i)} \in \mathbb{R}^m$ places this scalar into the i -th entry of y_t . The measurement noise is modeled as $v_t \sim \mathcal{N}(0, V)$, where $V = \text{diag}(\sigma_1^2, \dots, \sigma_m^2) \in \mathbb{R}^{m \times m}$ is the measurement noise covariance matrix.

B. Quadratic observation selection

Let $S \in \{0, 1\}^{k \times m}$ be a binary selection matrix that selects k out of m available measurements for use in the measurement update. As we consider a time-invariant selection, the same subset of measurements is used at every time step after selection. The selected measurement vector is denoted by $y_t(S) \in \mathbb{R}^k$. Each row of S contains exactly one nonzero entry, and each column contains at most one nonzero entry; hence, each selected row corresponds to one unique measurement channel, and no channel is selected more than once.

For nonlinear observation models, the posterior state distribution associated with a selected subset of measurements is generally intractable, making the exact estimation error covariance unavailable in closed form. Furthermore, linearization-based surrogates do not follow the true error covariance of the original nonlinear system, so guarantees established for the linearized model do not

directly extend to the nonlinear setting. To overcome this challenge, [19] proposed optimizing the Van Trees bound [21] for quadratic models, since this bound both preserves a direct link to estimation performance and admits a tractable closed-form characterization under quadratic observations.

Proposition 1. ([21], Van Trees regularity condition for selected quadratic observations)

Assume the Van Trees regularity condition in [19] holds:

$$\int \nabla_x \left(p(x) \mathbb{E}_{y|x} [\hat{x}(y(S)) - x] \right) dx = 0, \quad (2)$$

where $\mathbb{E}_{y|x}[\cdot]$ denotes expectation with respect to the conditional distribution of the selected measurements $y(S)$ given x . A sufficient condition is that the estimator is conditionally unbiased.

Proposition 2. ([21], Van Trees lower bound with selected quadratic observations)

Under assumption in Eq. (2), let $x \in \mathbb{R}^n$ be a random state vector with prior density $p(x)$, such that $\mathbb{E}[x] = 0$ and $\text{Cov}(x) = P$. Then, for the quadratic observation model, the Van Trees bound (B_S) admits the closed form

$$B_S := \left(I_x + \sum_{i \in S} \frac{1}{\sigma_i^2} \left(M^{(i)} P M^{(i)\top} + c_i c_i^\top \right) \right)^{-1}, \quad (3)$$

where

$$I_x = \mathbb{E}_x \left[\nabla_x \log p(x) \nabla_x \log p(x)^\top \right] \quad (4)$$

Here, $c_i \in \mathbb{R}^n$ denotes the transpose of the i -th row of the measurement matrix C . Moreover, the mean-square error matrix satisfies $\mathbb{E}[(x - \hat{x}_S)(x - \hat{x}_S)^\top] \succeq B_S$.

Problem. (Performance constrained minimum measurement selection.)

$$\begin{aligned} \min_{S \subseteq G} \quad & |S| \\ \text{subject to} \quad & h(S) \leq R \end{aligned} \quad (5)$$

where $h(S)$ is a set-function performance metric associated with the selected subset S . Here, we adopt the A-optimality criterion based on the Van Trees lower bound and define

$$h(S) = \text{Tr}(B_S).$$

Thus, the constraint $h(S) \leq R$ enforces a prescribed estimation-accuracy level, while the objective seeks the smallest subset of selected observations.

Although this criterion is motivated by the Van Trees lower bound on the MSE and therefore remains explicitly connected to estimation performance, it is still a surrogate rather than the exact MSE itself. Nonetheless, by *Proposition 2*, it provides a principled approximation that utilizes more informative observation subsets than generic heuristics. Furthermore, since the Van Trees bound is asymptotically tight, it becomes increasingly accurate and the selected subsets are expected to be near-optimal.

The above optimization problem is NP-hard, since solving it exactly requires a combinatorial search over all subsets

$S \subseteq G$. Consequently, exact optimization becomes intractable in large-scale settings, motivating the use of polynomial-time approximation algorithms. Greedy methods are particularly useful when the objective exhibits submodularity or supermodularity, as such structure enables approximation guarantees. However, A-optimality: $\text{Tr}(B_S)$ is generally neither submodular nor supermodular. To address this, we instead quantify the extent to which the objective approximately satisfies supermodularity through the supermodularity ratio.

IV. GREEDY MINIMIZATION AND GUARANTEES FOR APPROXIMATELY SUPERMODULAR SENSOR SELECTION

In this section, we briefly describe the greedy algorithm used to solve the performance-constrained observation selection problem, and then present the associated measurement-utilization guarantee based on the supermodularity ratio γ_h . Also, a lower bound on γ_h is derived for the proposed quadratic observation model.

Algorithm 1: Greedy measurement selection for Problem (5)

```

1 Input: Ground set  $G$ , performance threshold  $R$ , set
   function  $h(\cdot)$ 
2 Initialize:  $S \leftarrow \emptyset$ 
3 while  $h(S) > R$  do
   |  $j^* \leftarrow \arg \max_{j \in G \setminus S} [h(S) - h(S \cup \{j\})]$ 
   |  $S \leftarrow S \cup \{j^*\}$ 
return  $S$ 

```

Algorithm 1 starts from the empty set and iteratively adds measurements from the ground set G until the required performance level is achieved. At each iteration, the algorithm evaluates the marginal decrease $h(S) - h(S \cup \{j\})$, $j \in G \setminus S$ for all currently unselected measurements, and selects the measurement j^* that provides the largest reduction in the objective. The active set is then updated as $S \leftarrow S \cup \{j^*\}$. The procedure terminates once the performance constraint $h(S) \leq R$ is satisfied, and returns the selected subset S .

Proposition 3. ([22], One-step greedy progress)

Let G be a finite ground set, and let $h : 2^G \rightarrow \mathbb{R}$ be a nonincreasing set function. Let $S^* \subseteq G$ be an optimal subset with $|S^*| \geq 1$, and let $\{S_i\}_{i \geq 0}$ be a greedy sequence with $S_0 = \emptyset$ and $S_i = S_{i-1} \cup \{j_i\}$ for $i \geq 1$. If h has supermodularity ratio $\gamma_h \in (0, 1]$, then for every $i \geq 1$,

$$h(S_{i-1}) - h(S_i) \geq \frac{\gamma_h}{|S^*|} (h(S_{i-1}) - h(S^*)). \quad (6)$$

Proposition 4. (Geometric convergence of greedy)

Under the assumptions of Proposition 3, for any $\ell \geq 1$,

$$h(\emptyset) - h(S_{\ell-1}) \geq \left(1 - e^{-\gamma_h(\ell-1)/|S^*|} \right) (h(\emptyset) - h(S^*)). \quad (7)$$

Theorem 1. (Greedy measurement-utilization bound)

Let $S_{\ell-1}^g$ denote the greedy set at step $\ell - 1$, i.e., the last greedy iterate that still satisfies $h(S_{\ell-1}^g) > R$ and suppose

Algorithm 1 ends at step ℓ . Let S^* be an optimal solution to Problem (5). Then, using *Proposition 4.*, the inequality of measurement usage bound satisfies as below:

$$\frac{\ell}{|S^*|} \leq 1 + \frac{1}{\gamma_h} \log \left(\frac{h(\emptyset) - R}{h(S_{\ell-1}^g) - R} \right). \quad (8)$$

Proof of Theorem 1. From *Proposition 4*, evaluated at step $\ell - 1$, we have

$$h(\emptyset) - h(S_{\ell-1}^g) \geq \left(1 - e^{-\frac{\gamma_h}{|S^*|}(\ell-1)}\right) (h(\emptyset) - h(S^*)). \quad (9)$$

Define $\tau := 1 - e^{-\frac{\gamma_h}{|S^*|}(\ell-1)} \in (0, 1)$. Then the above inequality can be rearranged as

$$h(S_{\ell-1}^g) \leq (1 - \tau)h(\emptyset) + \tau h(S^*). \quad (10)$$

Since S^* is feasible, we have $h(S^*) \leq R$, and therefore

$$h(S_{\ell-1}^g) \leq (1 - \tau)h(\emptyset) + \tau R. \quad (11)$$

Because $S_{\ell-1}^g$ is the last iterate before termination, it still violates the target constraint, so $h(S_{\ell-1}^g) > R$. Let $h(S_{\ell-1}^g) = R + \delta$, $\delta := h(S_{\ell-1}^g) - R > 0$. Substituting this into Eq. (11) yields $R + \delta \leq (1 - \tau)h(\emptyset) + \tau R$, which simplifies to $\delta \leq (1 - \tau)(h(\emptyset) - R)$.

Using $1 - \tau = e^{-\frac{\gamma_h}{|S^*|}(\ell-1)}$, we obtain

$$h(S_{\ell-1}^g) - R \leq e^{-\frac{\gamma_h}{|S^*|}(\ell-1)} (h(\emptyset) - R). \quad (12)$$

Taking logarithms gives

$$\log \left(\frac{h(S_{\ell-1}^g) - R}{h(\emptyset) - R} \right) \leq -\frac{\gamma_h}{|S^*|}(\ell - 1), \quad (13)$$

or equivalently,

$$\frac{\ell - 1}{|S^*|} \leq \frac{1}{\gamma_h} \log \left(\frac{h(\emptyset) - R}{h(S_{\ell-1}^g) - R} \right). \quad (14)$$

As $|S^*| \geq 1$ satisfies, following equation holds:

$$\frac{\ell}{|S^*|} \leq 1 + \frac{1}{\gamma_h} \log \left(\frac{h(\emptyset) - R}{h(S_{\ell-1}^g) - R} \right). \quad (15)$$

This expression bounds to approximate how many selections for the optimal solution $|S^*|$ are needed to reach or exceed the target accuracy R , showing that the required number of steps grows logarithmically with the remaining normalized gap and is scaled by the inverse of supermodularity factor γ_h^{-1} .

Theorem 2. (Lower bound on the supermodularity ratio γ_h)

Consider *Proposition 2.* and the set function $h(S) = \text{Tr}(B_S)$ with $I_x \succ 0$, $P \succ 0$, $\sigma_i^2 > 0$, and $M^{(i)} \in \mathbb{R}^{n \times n}$ invertible for all $i \in G$. For any $S \subseteq G$, define $F_S := I_x + \sum_{i \in S} I_i$, $\tilde{F}_{S,j} := F_S + \sigma_j^2 I_j$ and let $\tilde{F}_G := I_x + \sum_{i \in G} I_i$. Then *Definition 3* admits the lower bound:

$$\gamma_h \geq \min\{\underline{\gamma}_f, \underline{\gamma}_g\}, \quad (16)$$

where

$$\underline{\gamma}_f := \frac{\lambda_{\min}(I_x)}{\lambda_{\max}(\tilde{F}_G)} \cdot \frac{1 + \sigma_j^2 \lambda_{\min}(I_x) / \|c_j\|^2}{1 + \sigma_j^2 \lambda_{\max}(\tilde{F}_G) / \|c_j\|^2}, \quad (17)$$

and

$$\underline{\gamma}_g := \frac{\lambda_{\min}(I_x)^2}{\lambda_{\max}(\tilde{F}_G)^2} \cdot \frac{\text{Tr}((V_j + I_x^{-1})^{-1})}{\text{Tr}((V_j + \tilde{F}_G^{-1})^{-1})}, \quad (18)$$

with

$$V_j := (M^{(j)\top})^{-1} \sigma_j^2 P^{-1} (M^{(j)})^{-1}.$$

Consequently, h is approximately supermodular with ratio at least $\min\{\underline{\gamma}_f, \underline{\gamma}_g\}$.

Proof of Theorem 2. Let any $S \subseteq T \subseteq G$ and $j \in G \setminus T$. Define the marginal decrease

$$\Delta_j(S) := h(S) - h(S \cup \{j\}), \quad \Delta_j(T) := h(T) - h(T \cup \{j\}).$$

Using *Lemma 7*, with the marginal decrease can be decomposed [20] as $\Delta_j(S) = f_j(S) + g_j(S)$ where

$$f_j(S) := \frac{c_j^\top \tilde{F}_{S,j}^{-2} c_j}{\sigma_j^2 + c_j^\top \tilde{F}_{S,j}^{-1} c_j},$$

and

$$g_j(S) := \text{Tr}(F_S^{-1} M^{(j)} (\sigma_j^2 P^{-1} + M^{(j)\top} F_S^{-1} M^{(j)})^{-1} M^{(j)\top} F_S^{-1}).$$

Similarly, $\Delta_j(T) = f_j(T) + g_j(T)$.

Since $c_j \neq 0$, full rank $M^{(j)}$, $F_S \succ 0$, and $\tilde{F}_{S,j} \succ 0$, all four quantities $f_j(S)$, $f_j(T)$, $g_j(S)$, $g_j(T)$ are strictly positive. Hence by *Lemma 4*,

$$\frac{\Delta_j(S)}{\Delta_j(T)} = \frac{f_j(S) + g_j(S)}{f_j(T) + g_j(T)} \geq \min \left\{ \frac{f_j(S)}{f_j(T)}, \frac{g_j(S)}{g_j(T)} \right\}. \quad (19)$$

We first bound the f_j -ratio. Define $\alpha_S := c_j^\top \tilde{F}_{S,j}^{-1} c_j$, $\alpha_T := c_j^\top \tilde{F}_{T,j}^{-1} c_j$, $\beta_S := c_j^\top \tilde{F}_{S,j}^{-2} c_j$, $\beta_T := c_j^\top \tilde{F}_{T,j}^{-2} c_j$. Then

$$\frac{f_j(S)}{f_j(T)} = \frac{\beta_S}{\beta_T} \cdot \frac{\sigma_j^2 + \alpha_T}{\sigma_j^2 + \alpha_S}.$$

Using *Lemma 5*, $\beta_T \leq \lambda_{\max}(\tilde{F}_{T,j}^{-1}) \alpha_T$, $\beta_S \geq \lambda_{\min}(\tilde{F}_{S,j}^{-1}) \alpha_S$. Therefore,

$$\frac{f_j(S)}{f_j(T)} \geq \frac{\lambda_{\min}(\tilde{F}_{S,j}^{-1})}{\lambda_{\max}(\tilde{F}_{T,j}^{-1})} \cdot \frac{\alpha_S}{\alpha_T} \cdot \frac{\sigma_j^2 + \alpha_T}{\sigma_j^2 + \alpha_S} = \frac{\lambda_{\min}(\tilde{F}_{S,j}^{-1})}{\lambda_{\max}(\tilde{F}_{T,j}^{-1})} \cdot \frac{\sigma_j^2 / \alpha_T + 1}{\sigma_j^2 / \alpha_S + 1}.$$

Since $S \subseteq T \subseteq G$, we have $I_x \preceq \tilde{F}_{S,j} \preceq \tilde{F}_{T,j} \preceq \tilde{F}_G$. Using *Lemma 2,3*,

$$\frac{\lambda_{\min}(\tilde{F}_{S,j}^{-1})}{\lambda_{\max}(\tilde{F}_{T,j}^{-1})} = \frac{\lambda_{\min}(\tilde{F}_{T,j})}{\lambda_{\max}(\tilde{F}_{S,j})} \geq \frac{\lambda_{\min}(I_x)}{\lambda_{\max}(\tilde{F}_G)}.$$

Moreover, by *Lemma 5*,

$$\alpha_T \leq \lambda_{\max}(\tilde{F}_{T,j}^{-1}) \|c_j\|^2 \leq \lambda_{\max}(I_x^{-1}) \|c_j\|^2 = \frac{\|c_j\|^2}{\lambda_{\min}(I_x)},$$

and

$$\alpha_S \geq \lambda_{\min}(\tilde{F}_{S,j}^{-1}) \|c_j\|^2 \geq \lambda_{\min}(\tilde{F}_G^{-1}) \|c_j\|^2 = \frac{\|c_j\|^2}{\lambda_{\max}(\tilde{F}_G)}.$$

Substituting these bounds yields

$$\frac{f_j(S)}{f_j(T)} \geq \frac{\lambda_{\min}(I_x)}{\lambda_{\max}(\tilde{F}_G)} \cdot \frac{1 + \sigma_j^2 \lambda_{\min}(I_x) / \|c_j\|^2}{1 + \sigma_j^2 \lambda_{\max}(\tilde{F}_G) / \|c_j\|^2} = \gamma_f. \quad (20)$$

Next, we bound the g_j -ratio. Using *Lemma 1*,

$$g_j(S) = \text{Tr} \left(M^{(j)\top} F_S^{-2} M^{(j)} (\sigma_j^2 P^{-1} + M^{(j)\top} F_S^{-1} M^{(j)})^{-1} \right).$$

Since $M^{(j)}$ is invertible,

$$g_j(S) = \text{Tr} \left(F_S^{-2} ((M^{(j)\top})^{-1} \sigma_j^2 P^{-1} (M^{(j)})^{-1} + F_S^{-1})^{-1} \right).$$

Define $V_j := (M^{(j)\top})^{-1} \sigma_j^2 P^{-1} (M^{(j)})^{-1}$, $U_S := V_j + F_S^{-1}$, $U_T := V_j + F_T^{-1}$. Then $g_j(S) = \text{Tr}(F_S^{-2} U_S^{-1})$, $g_j(T) = \text{Tr}(F_T^{-2} U_T^{-1})$.

By *Lemma 6*,

$$g_j(S) \geq \lambda_{\min}(F_S^{-2}) \text{Tr}(U_S^{-1}), \quad g_j(T) \leq \lambda_{\max}(F_T^{-2}) \text{Tr}(U_T^{-1}),$$

hence

$$\frac{g_j(S)}{g_j(T)} \geq \frac{\lambda_{\min}(F_S^{-2})}{\lambda_{\max}(F_T^{-2})} \cdot \frac{\text{Tr}(U_S^{-1})}{\text{Tr}(U_T^{-1})}.$$

Since $I_x \preceq F_S \preceq F_T \preceq \tilde{F}_G$, *Lemma 2,3* imply

$$\lambda_{\min}(F_S^{-2}) \geq \frac{1}{\lambda_{\max}(\tilde{F}_G)^2}, \quad \lambda_{\max}(F_T^{-2}) \leq \frac{1}{\lambda_{\min}(I_x)^2}.$$

Also, $\tilde{F}_G^{-1} \preceq F_T^{-1} \preceq F_S^{-1} \preceq I_x^{-1}$, so $V_j + \tilde{F}_G^{-1} \preceq U_T \preceq U_S \preceq V_j + I_x^{-1}$.

Applying *Lemma 2* again,

$$(V_j + I_x^{-1})^{-1} \preceq U_S^{-1} \preceq U_T^{-1} \preceq (V_j + \tilde{F}_G^{-1})^{-1}.$$

Taking traces yields

$$\text{Tr}(U_S^{-1}) \geq \text{Tr}((V_j + I_x^{-1})^{-1}), \quad \text{Tr}(U_T^{-1}) \leq \text{Tr}((V_j + \tilde{F}_G^{-1})^{-1}).$$

Therefore,

$$\frac{g_j(S)}{g_j(T)} \geq \frac{\lambda_{\min}(I_x)^2}{\lambda_{\max}(\tilde{F}_G)^2} \cdot \frac{\text{Tr}((V_j + I_x^{-1})^{-1})}{\text{Tr}((V_j + \tilde{F}_G^{-1})^{-1})} = \gamma_g. \quad (21)$$

Combining (19), (20), and (21), we obtain

$$\frac{\Delta_j(S)}{\Delta_j(T)} \geq \min\{\gamma_f, \gamma_g\}.$$

Since this holds for every $S \subseteq T \subseteq G$ and every $j \in G \setminus T$, taking the minimum over all such triples gives

$$\gamma_h \geq \min\{\gamma_f, \gamma_g\}.$$

This result provides an explicit lower bound on the supermodularity ratio γ_h for a substantially broader class of quadratic observation models than prior work. In particular, the bound remains valid when the linear measurement term is present, i.e., $c_i \neq 0$, and when the quadratic factor $M^{(i)}$ is full-rank rather than rank-1. This is important because many practical sensing models contain both linear and quadratic components, and their quadratic structure is often not restricted to a rank-1 form.

V. SIMULATION SETUP AND EVALUATION

To empirically validate the proposed theory, we evaluate the performance-constrained measurement selection framework on synthetic quadratic observation models and compare three selection strategies: (i) brute-force optimal selection for Problem (5), (ii) greedy selection using the proposed Van Trees-based quadratic objective, and (iii) greedy selection based on a linearized approximation of the observation model. The simulations are designed to validate two main claims of the paper: first, that greedy selection under the proposed quadratic model achieves near-optimal measurement utilization relative to brute force; and second, that the lower bound on the supermodularity ratio remains valid in broader settings with nonzero linear terms and full-rank quadratic sensing matrices.

We generate quadratic observation models of the form in (1), with measurement channels parameterized by linear terms c_i and quadratic factors $M^{(i)}$. For each trial, we construct the Van Trees A-optimality metric $h(S) = (B_S)$ from Proposition 2 and solve the performance-constrained minimum-measurement problem in (5). The greedy quadratic method applies Algorithm 1 directly to this objective, while the linearized baseline applies greedy selection to a locally linearized surrogate of the same observation model. Since exact optimization over all subsets becomes combinatorial, brute-force search is used only for moderate problem sizes, which is sufficient to provide the optimal subset size $|S^*|$ needed for comparison with the greedy methods.

Fig. 1 compares sensor utilization across different target accuracy levels. The horizontal axis shows the target accuracy ratio, and the vertical axis reports the utilization ratio $\ell/|S^*|$, where ℓ is the number of measurements selected by the corresponding greedy method and $|S^*|$ is the optimal subset size. The dashed green line denotes the brute-force optimum, which is identically one after normalization. The blue curve corresponds to the proposed greedy quadratic method and remains close to one across the entire range, indicating that the proposed method selects nearly the same number of measurements as the optimal solution. In contrast, the red curve corresponds to greedy selection under linearization and shows a substantially larger utilization ratio, with the gap increasing as the target accuracy becomes more stringent. This behavior is consistent with the discussion in Section III-B: linearization-based surrogates do not faithfully track the original quadratic estimation objective, whereas the proposed Van Trees surrogate remains explicitly tied to the nonlinear model.

To directly evaluate the supermodularity-ratio analysis in Theorem 2, Fig. 2 compares the empirical supermodularity ratio with the proposed lower bound under two parameter sweeps. In the left panel, we vary the scale of the linear measurement term C , which controls the departure from the pure-quadratic setting studied in prior work. In the right panel, we vary the measurement-noise scale σ^2 . In both cases, the blue curve shows the empirical ratio computed from marginal decreases of $h(S)$, while the orange curve

shows the proposed lower bound $\min\{\gamma_f, \gamma_g\}$. For reference, the green curve shows the bound derived from the prior restricted quadratic setting [19], plotted only as a comparison baseline. As shown in Fig. 2, the proposed bound remains below the empirical ratio across the tested parameter range, as expected for a valid conservative certificate. More importantly, the left panel demonstrates that the proposed lower bound continues to apply when the linear term is nonzero, whereas the prior bound is tied to the pure-quadratic regime and does not capture this broader mixed linear–quadratic setting. The right panel further shows that the bound tracks the correct trend as the measurement noise level changes, supporting the robustness of the proposed analysis.

Overall, the simulations support both the algorithmic and theoretical claims of the paper. Fig. 1 shows that greedy selection based on the proposed quadratic objective remains close to brute-force optimal in terms of total measurement usage, while significantly outperforming a linearization-based greedy baseline. Fig. 2 shows that the proposed supermodularity-ratio lower bound is numerically valid and remains applicable in broader regimes with nonzero linear terms and full-rank quadratic factors. These results confirm that the proposed framework provides a practically meaningful and theoretically grounded approximation for resource-aware measurement selection under quadratic observations.

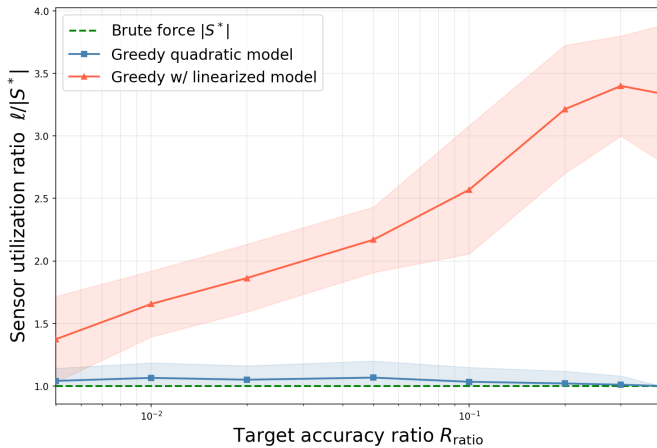


Fig. 1: Sensor utilization ratio versus target accuracy ratio. The proposed greedy quadratic method remains close to the brute-force optimum across the tested range, while the linearized greedy baseline requires substantially more measurements, especially at stricter accuracy targets.

VI. CONCLUSIONS

We proposed a resource-aware framework for measurement selection under nonlinear quadratic observation models that minimizes the number of selected measurements while meeting a target estimation-accuracy requirement. By leveraging greedy selection with a Van Trees-based A-optimality objective, we established a measurement-utilization guarantee relative to the optimal subset size and derived an explicit lower bound on the supermodularity ratio. Unlike prior analyses under restrictive pure-quadratic or low-rank settings, our result applies to a broader class of sensing models with nonzero linear terms and full-rank quadratic

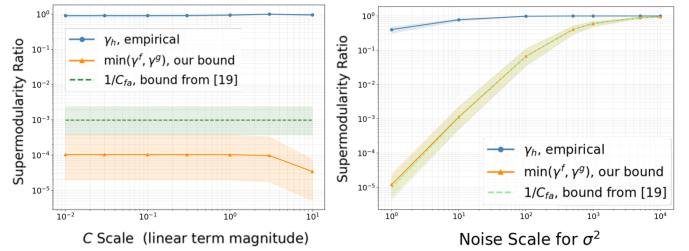


Fig. 2: Empirical supermodularity ratio and theoretical lower bounds under parameter sweeps. Left: varying the linear-term scale of C . Right: varying the measurement-noise scale σ^2 . The proposed bound $\min\{\gamma_f, \gamma_g\}$ remains valid across both sweeps and applies beyond the restrictive pure-quadratic setting considered in prior work.

sensing matrices. Future work will extend this analysis to time-varying scheduling and other non-submodular estimation metrics.

REFERENCES

- [1] Liu, Jun, and Ryan K. Williams. "Optimal intermittent deployment and sensor selection for environmental sensing with multi-robot teams." 2018 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2018.
- [2] Mourikis, Anastasios I., and Stergios I. Roumeliotis. "Optimal sensor scheduling for resource-constrained localization of mobile robot formations." IEEE Transactions on Robotics 22.5 (2006): 917-931.
- [3] Golovin, Daniel, Matthew Faulkner, and Andreas Krause. "Online distributed sensor selection." Proceedings of the 9th ACM/IEEE International Conference on Information Processing in Sensor Networks. 2010.
- [4] Thattai, Gautam, and Urbashi Mitra. "Sensor selection and power allocation for distributed estimation in sensor networks: Beyond the star topology." IEEE Transactions on Signal Processing 56.7 (2008): 2649-2661.
- [5] Lovisari, Enrico, Carlos Canudas de Wit, and Alain Y. Kibangou. "Optimal sensor placement in road transportation networks using virtual variances." 2015 54th IEEE Conference on Decision and Control (CDC). IEEE, 2015.
- [6] He, Ying, and K. P. Chong. "Sensor scheduling for target tracking in sensor networks." 2004 43rd IEEE Conference on Decision and Control (CDC)(IEEE Cat. No. 04CH37601). Vol. 1. IEEE, 2004.
- [7] Cao, Nianxia, et al. "Sensor selection for target tracking in wireless sensor networks with uncertainty." IEEE Transactions on signal Processing 64.20 (2016): 5191-5204.
- [8] Subramanian, Ramanan, and Faramarz Fekri. "Sleep scheduling and lifetime maximization in sensor networks: fundamental limits and optimal solutions." Proceedings of the 5th international conference on Information processing in sensor networks. 2006.
- [9] Tzoumas, Vasileios, Ali Jadbabaie, and George J. Pappas. "Sensor placement for optimal Kalman filtering: Fundamental limits, submodularity, and algorithms." 2016 American control conference (ACC). IEEE, 2016.
- [10] Zhang, Haotian, Raid Ayoub, and Shreyas Sundaram. "Sensor selection for Kalman filtering of linear dynamical systems: Complexity, limitations and greedy algorithms." Automatica 78 (2017): 202-210.
- [11] Ye, Lintao, Sandip Roy, and Shreyas Sundaram. "On the complexity and approximability of optimal sensor selection for Kalman filtering." 2018 Annual American Control Conference (ACC). IEEE, 2018.
- [12] Shamaiah, Manohar, Siddhartha Banerjee, and Haris Vikalo. "Greedy sensor selection: Leveraging submodularity." 49th IEEE conference on decision and control (CDC). IEEE, 2010.
- [13] Nemhauser, George L., Laurence A. Wolsey, and Marshall L. Fisher. "An analysis of approximations for maximizing submodular set functions—I." Mathematical programming 14 (1978): 265-294.
- [14] Fisher, Marshall L., George L. Nemhauser, and Laurence A. Wolsey. "An analysis of approximations for maximizing submodular set functions—II." Polyhedral Combinatorics: Dedicated to the memory of DR Fulkerson. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009. 73-87.

- [15] Tzoumas, Vasileios, et al. "LQG control and sensing co-design." *IEEE Transactions on Automatic Control* 66.4 (2020): 1468-1483.
- [16] Summers, Tyler H., Fabrizio L. Cortesi, and John Lygeros. "On submodularity and controllability in complex dynamical networks." *IEEE Transactions on Control of Network Systems* 3.1 (2015): 91-101.
- [17] Shen, Xiaojing, Sijia Liu, and Pramod K. Varshney. "Sensor selection for nonlinear systems in large sensor networks." *IEEE Transactions on Aerospace and Electronic Systems* 50.4 (2014): 2664-2678.
- [18] Haber, Aleksandar, Ferenc Molnar, and Adilson E. Motter. "State observation and sensor selection for nonlinear networks." *IEEE Transactions on Control of Network Systems* 5.2 (2017): 694-708.
- [19] Hashemi, Abolfazl, et al. "Submodular observation selection and information gathering for quadratic models." *International Conference on Machine Learning*. PMLR, 2019.
- [20] Hashemi, Abolfazl, et al. "Supplementary material for "Submodular Observation Selection and Information Gathering for Quadratic Models"."
- [21] Van Trees, Harry L., and Kristine L. Bell. *Detection estimation and modulation theory, part I: detection, estimation, and filtering theory*. John Wiley Sons, 2013.
- [22] Krause, Andreas, and Carlos Guestrin. "A note on the budgeted maximization of submodular functions." (2005): 103.